

Python Assignment – Week 14

File Handling & Functions

Objective:

This hands-on assignment will help learners understand **file handling, functions, conditional statements, and error handling** in Python. By the end, students will be able to **open, read, write, and append files**, use **functions for reusability**, and apply **try-except for error handling**.

Assignment Tasks:

1. Read from a File (10 Points)

Problem Statement:

Write a Python program to **read** a text file and display its contents.

☒ Steps:

- Open a file named "sample.txt" in read mode.
- Read and print the content of the file.
- Ensure the file closes properly after reading.

Example Output:

```
Contents of the file:
Hello, this is a sample text file.
This file is used to demonstrate file handling in Python.
```

Solution

1. Read from a File Solution

```
# Reading a file
with open("sample.txt", "r") as file:
    content = file.read()
    print("Contents of the file:\n", content)
```

2. Write and Append to a File (10 Points)

Problem Statement:

Write a Python program that **writes user input** into a file and then **appends additional data**.

☑ Steps:

- Create a new file "output.txt" and write "This is the first line.\n"
- Append "This is an appended line.\n" to the same file.
- Read and display the file's contents.

🚩 Example Output:

```
Contents of output.txt:
This is the first line.
This is an appended line.
```

Solution

```
# Writing and Appending to a File
with open("output.txt", "w") as file:
    file.write("This is the first line.\n")

with open("output.txt", "a") as file:
    file.write("This is an appended line.\n")

# Reading and Displaying the File Content
with open("output.txt", "r") as file:
    print("Contents of output.txt:\n", file.read())
```

3. Function to Count Words in a File (15 Points)

Problem Statement:

Write a function that takes a **filename** as an argument and returns the **total number of words** in that file.

☑ Steps:

- Open the file in read mode.
- Count the total number of words in the file.
- Return and display the word count.

🚩 Example Output:

```
Total words in sample.txt: 15
```

Solution

```
# Function to Count Words in a File
def count_words(filename):
    with open(filename, "r") as file:
        words = file.read().split()
        return len(words)

# Example usage
```

```
word_count = count_words("sample.txt")
print(f"Total words in sample.txt: {word_count}")
```

4. Try-Except for Error Handling (15 Points)

Problem Statement:

Write a Python program that attempts to open a **non-existent file** and handles the error gracefully using a **try-except** block.

☒ Steps:

- Try opening "missing_file.txt".
- If the file doesn't exist, display "File not found. Please check the filename."

Example Output:

```
File not found. Please check the filename.
```

Solution

```
# Handling File Not Found Error
```

```
try:
```

```
    with open("missing_file.txt", "r") as file:
```

```
        print(file.read())
```

```
except FileNotFoundError:
```

```
    print("File not found. Please check the filename.")
```